

Anaconda & VS Code

Complete Setup Guide for Python & Jupyter Notebooks

What is Anaconda?

All-in-One Python Distribution

Includes Python, Jupyter Notebook, and 250+ pre-installed packages

Perfect for Data Science & AI

No need to install packages individually - saves hours of setup time

Environment Management

Create isolated environments for different projects easily

Step 1: Download Anaconda

1

Visit Official Website

www.anaconda.com/download

Click on 'Free Download' button

2

Choose Your OS

- Windows (64-bit)
- macOS
- Linux

3

Download Size

~600 MB - may take 5-15 minutes



Important Notes

- ✓ Download the latest version (Python 3.11+)
- ✓ Choose the Individual Edition (Free)
- ✓ Stable internet connection recommended
- ✓ Requires ~3 GB free disk space

Step 2: Install Anaconda

Windows Installation Steps:

1. Run the downloaded .exe file (Right-click → Run as Administrator)
2. Click 'Next' on the welcome screen
3. Read and accept the License Agreement
4. Select 'Just Me' (recommended) installation type
5. Choose installation location (default is fine: C:\Users\YourName\anaconda3)
6. **⚠ IMPORTANT: Check 'Add Anaconda to PATH' (optional but recommended)**
7. Click 'Install' - this will take 10-15 minutes
8. Click 'Finish' when installation completes

Step 3: Verify Installation

Open Anaconda Prompt

Windows:

Start Menu → Anaconda3 →
Anaconda Prompt

Mac/Linux:

Open Terminal application

Check Python Version

```
python --version
```

Expected Output

```
Python 3.11.5
```

✓ Success Indicators:

- Python version 3.11 or higher
- No error messages
- Command executes successfully

If Installation Failed:

Reinstall Anaconda and ensure 'Add to PATH' is checked

Step 4: Download VS Code

Download Process

1. Visit Official Site

code.visualstudio.com

2. Click 'Download'

Choose your operating system:

- Windows (User Installer, 64-bit)
- macOS
- Linux (.deb or .rpm)

3. File Size

~90 MB - quick download

Why VS Code?

✓ Lightweight & Fast

Opens instantly, low system resources

✓ Excellent Jupyter Support

Run .ipynb files directly in editor

✓ Powerful Extensions

Python, Jupyter, IntelliSense support

✓ Industry Standard

Used by professional developers worldwide

Step 5: Install VS Code

Installation Steps (Windows):

1. Run the downloaded installer (.exe file)
2. Accept the License Agreement
3. Choose installation location (default is recommended)
4. ✓ Check 'Add to PATH' (important!)
5. ✓ Check 'Create a desktop icon'
6. ✓ Check 'Register Code as an editor for supported file types'
7. Click 'Install' - takes about 2-3 minutes
8. Click 'Finish' and launch VS Code

Step 6: Install Python Extension

In VS Code:

1. Open Extensions

Click the Extensions icon (or Ctrl+Shift+X)

2. Search for 'Python'

Type 'Python' in search bar

3. Install Python Extension

Publisher: Microsoft

Click 'Install' button

4. Install Jupyter Extension

Search 'Jupyter' and install

Also by Microsoft

✓ Required Extensions

1. Python

(by Microsoft)

2. Jupyter

(by Microsoft)

Recommended (Optional)

- Pylance (Better IntelliSense)
- Python Indent
- autoDocstring

Step 7: Link Anaconda to VS Code

Select Python Interpreter:

1. Open VS Code
2. Press Ctrl+Shift+P (Windows) or Cmd+Shift+P (Mac)

Opens Command Palette

3. Type: Python: Select Interpreter
4. Choose Anaconda's Python from the list:

Look for: Python 3.11.x ('base': conda)

Path: C:\Users\YourName\anaconda3\python.exe

5. ✓ **Anaconda is now linked!**

You'll see the Python version in bottom-left corner

Step 8: Test Python File

Create a Python File:

1. Create new folder for practice
2. Open folder in VS Code
3. Create new file: test.py
4. Write code:

```
# My first Python program  
print("Hello, AI ML World!")  
print("Python is working!")
```

5. Run: Click ► button or press F5

✓ Expected Output:

```
Hello, AI ML World!  
Python is working!
```

Success Indicators:

- ✓ Code executes without errors
- ✓ Output appears in Terminal
- ✓ Python version shown in terminal

Tip: Save files before running!
Use Ctrl+S (Windows) or Cmd+S (Mac)

Step 9: Test Jupyter Notebook

Create Jupyter Notebook:

1. Create new file: test.ipynb
2. VS Code will open Notebook interface
3. Select Anaconda kernel:
Click 'Select Kernel' (top-right)
Choose: Python 3.11.x ('base': conda)
4. Write code in first cell:

```
print("Jupyter is working!")  
2 + 2
```

5. Run cell: Click ▶ or Shift+Enter

✓ Expected Output:

```
Jupyter is working!  
4
```

Notebook Features:

- ✓ Run code in cells
- ✓ See output immediately
- ✓ Mix code with text (Markdown)
- ✓ Edit and re-run cells easily

**Jupyter is perfect for data analysis
and machine learning experiments!**

Managing Conda Environments

What are Environments?

Isolated spaces for different projects with their own packages and Python versions.

Essential Commands (Anaconda Prompt):

```
conda --version
```

Check if conda is installed

```
conda env list
```

List all environments

```
conda create -n myenv python=3.11
```

Create new environment

```
conda activate myenv
```

Activate an environment

```
conda deactivate
```

Deactivate current environment

```
conda remove -n myenv --all
```

Delete an environment

Installing Libraries (Future)

Two Methods:

Method 1: Using conda

```
conda install num py  
conda install pandas  
conda install matplotlib
```

Method 2: Using pip

```
pip install num py  
pip install pandas  
pip install matplotlib
```

Pre-installed in Anaconda:

- ✓ NumPy - Arrays & Math
- ✓ Pandas - Data Analysis
- ✓ Matplotlib - Plotting
- ✓ SciPy - Scientific Computing

Best Practices:

- Prefer conda over pip when possible
- Install in activated environment
- Check if already installed first

Common Issues & Solutions

Python command not found

- ✓ Reinstall Anaconda with 'Add to PATH' checked
OR restart computer after installation

VS Code doesn't detect Anaconda

- ✓ Use Ctrl+Shift+P → Python: Select Interpreter
Manually browse to anaconda3/python.exe

Jupyter kernel not starting

- ✓ Install/Reinstall Jupyter extension
Restart VS Code

'conda' not recognized

- ✓ Open Anaconda Prompt (not regular Command Prompt)
Or add Anaconda to system PATH

Environment activation fails

- ✓ Use: conda activate base first
Then create/activate your environment

✓ Final Verification Checklist

Software Installed:

- ☐ Anaconda installed successfully
- ☐ Python version 3.11+
- ☐ VS Code installed
- ☐ Python extension installed
- ☐ Jupyter extension installed

Functionality Tested:

- ☐ Python script runs (.py file)
- ☐ Jupyter notebook runs (.ipynb file)
- ☐ Anaconda interpreter selected
- ☐ Code executes without errors
- ☐ Output displays correctly
- ☐ Can create new environments

If all checked, you're ready to start coding!

You're All Set!

What You've Accomplished:

- ✓ Installed Anaconda (Python + 250+ packages)
- ✓ Installed VS Code (Professional code editor)
 - ✓ Linked Anaconda with VS Code
- ✓ Tested Python scripts and Jupyter notebooks
 - ✓ Learned environment management basics

Ready to Start Your AI/ML Journey!